

CHITRADURGA, India

WMO: 432330

Lat: 14.233N

Lon: 76.433E

Elev: 2405

StdP: 13.46

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	60.2	62.5	34.5	31.9	84.6	39.2	38.5	83.2	12.9	75.8	11.7	75.2	1.0	90	0.270

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	21.0	98.0	69.4	96.3	69.5	94.7	69.6	77.9	87.8	76.5	86.0	75.6	84.6	4.0	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.3	145.4	83.0	74.1	139.2	81.4	73.2	135.0	80.2	43.4	88.0	42.0	86.0	41.0	84.6	85.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.8	12.5	11.1	54.2	100.3	2.7	2.3	52.3	101.9	50.8	103.3	49.3	104.6	47.3	106.2		
(5)			49.1	80.0	2.0	2.1	47.6	81.5	46.5	82.7	45.4	83.9	44.0	85.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	78.0	74.2	78.7	83.4	85.1	83.2	78.5	76.3	76.0	76.6	76.8	74.5	73.2
	DBStd	4.50	2.66	2.91	2.62	2.56	2.85	2.66	2.11	1.94	1.91	2.22	2.59	2.41
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	10231	752	804	1035	1054	1030	855	815	805	798	831	734	718
	CDD65	4756	287	384	570	604	565	405	350	340	348	366	284	253
	CDH74	40524	2169	3783	6890	7624	6332	2805	1625	1505	1950	2453	1791	1597
	CDH80	16237	617	1666	3594	4118	3087	845	298	240	415	677	371	309
(14) Wind	WSAvg	4.7	3.1	3.3	3.4	3.9	5.8	7.5	7.4	6.8	5.2	3.0	3.0	3.4
(15) Precipitation	PrecAvg	23.9	0.0	0.0	0.2	1.8	3.0	2.1	3.1	2.8	4.3	4.8	1.6	0.5
	PrecMax	39.1	0.2	0.7	1.3	5.7	6.9	6.2	8.1	6.2	11.9	11.1	5.3	2.2
	PrecMin	14.8	0.0	0.0	0.0	0.0	0.2	0.2	0.8	0.7	0.6	0.0	0.0	0.0
	PrecStd	5.9	0.0	0.2	0.3	1.3	1.8	1.3	1.8	1.6	2.6	2.6	1.6	0.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	89.4	94.9	98.8	100.9	99.1	92.7	87.4	86.4	88.4	89.7	86.9	87.3
		MCWB	65.8	65.7	67.2	69.9	71.2	74.1	72.8	73.9	72.7	69.2	68.0	65.6
	2%	DB	87.1	92.5	96.8	98.7	96.6	89.4	85.3	84.5	86.2	87.3	85.2	84.9
		MCWB	65.3	66.3	67.4	69.3	71.9	73.8	72.7	73.6	72.4	69.3	68.2	65.9
	5%	DB	85.3	90.7	95.2	97.0	94.6	87.2	83.4	82.7	84.3	85.8	83.8	83.2
		MCWB	64.9	65.7	67.4	69.6	72.3	73.7	72.6	73.0	72.3	69.7	68.3	65.7
	10%	DB	83.6	88.8	93.7	95.2	92.5	84.9	81.4	81.0	82.6	83.9	82.3	81.6
		MCWB	64.6	65.3	67.5	70.2	72.9	73.3	72.5	72.6	72.2	70.5	68.0	65.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	72.1	74.3	78.6	80.9	79.0	78.0	76.3	76.4	76.5	76.1	74.5	72.9
		MCDB	81.3	87.5	92.6	92.2	88.8	85.5	83.6	83.3	82.7	82.1	81.0	81.0
	2%	WB	70.2	71.7	75.5	78.1	77.3	76.2	74.6	74.6	75.2	74.6	73.0	71.1
		MCDB	80.1	84.0	87.0	89.1	87.5	84.3	81.1	81.2	81.9	81.2	79.6	78.4
	5%	WB	68.8	70.1	73.7	76.2	76.3	75.1	73.7	73.7	74.2	73.7	72.0	70.0
		MCDB	79.3	82.4	85.6	86.6	86.3	83.4	79.9	80.0	80.7	80.5	78.6	77.2
	10%	WB	67.5	68.8	72.4	74.8	75.4	74.2	72.9	73.0	73.4	72.9	71.0	68.9
		MCDB	78.4	81.1	83.8	84.7	85.2	82.0	78.7	78.9	79.7	79.5	77.5	76.1
(35) Mean Daily Temperature Range	5% DB	MDBR	19.5	20.4	20.9	21.0	18.5	12.6	10.2	10.2	12.1	14.1	15.8	17.9
		MCDBR	21.1	22.0	22.3	22.6	21.1	16.0	13.7	13.0	15.0	17.7	18.2	19.9
		MCWBR	8.9	9.8	10.0	9.6	6.8	5.2	4.5	4.8	5.2	6.3	7.4	8.2
	5% WB	MCDBR	18.5	19.9	20.3	21.2	18.8	13.7	11.0	11.2	12.5	13.1	13.3	15.9
		MCWBR	8.8	9.8	10.0	9.6	7.4	5.8	4.9	5.1	5.9	6.3	7.8	
(40) Clear-Sky Solar Irradiance	taub		0.392	0.414	0.481	0.530	0.535	0.479	0.489	0.471	0.449	0.485	0.433	0.409
	taud		2.261	2.212	2.037	1.949	1.964	2.176	2.141	2.204	2.266	2.127	2.237	2.251
	Ebn at Noon		285	284	267	253	247	259	257	265	272	259	271	277
	Edh at Noon		42	46	56	61	59	47	49	47	44	49	43	42
(44) All-Sky Solar Radiation	RadAvg		1720	1957	2094	2131	2036	1624	1422	1462	1599	1608	1545	1574
	RadStd		69	76	72	56	88	121	96	70	112	112	113	81

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	+0.58	N/A	N/A	+0.94	N/A	N/A	N/A	N/A	+308	+310

Nomenclature: See separate page